

Mathematics

Rational functions

Grade 12 SE

I. Definition

- A function f is said to be a rational function if it is defined by $f(x) = \frac{P(x)}{q(x)}$ where P and q are two polynomials functions.
- That is ,rational functions are fractions with polynomials in the numerator and denominator

Examples

- The function f defined by $f(x) = \frac{2x+1}{x-1}$ is a rational function .
- The function g defined by $g(x) = \frac{x^2+2x-3}{2x-1}$ is a rational function .
- The function h defined by $h(x) = \frac{x+4}{2x^2+x-3}$ is a rational function

II. Domain of definition of a rational function .

- A rational function f , where $f(x) = \frac{P(x)}{q(x)}$ is defined for the values of x for which $q(x) \neq 0$

- Example

The function f defined by $f(x) = \frac{2x+1}{x-1}$ is defined when

- $x - 1 \neq 0$
 $x \neq 1.$

Then the domain of definition of f is $D_f =]-\infty; 1[\cup]1; +\infty[$

Examples

- The function f defined by $f(x) = \frac{2x+1}{x-1}$ is defined when
 $x - 1 \neq 0$
 $x \neq 1$.
- Then the domain of definition of f is $D_f =]-\infty; 1[\cup]1; +\infty[$
- The function h defined by $h(x) = \frac{x+4}{2x^2+x-3}$ is defined when $2x^2+x-3 \neq 0$.
 $x \neq 1$ and $x \neq -\frac{3}{2}$
- $D_h =]-\infty; -\frac{3}{2}[\cup]-\frac{3}{2}; 1[\cup]1; +\infty[$

Examples

- The function g defined by $g(x) = \frac{x+4}{x^2+1}$ is defined when $x^2+1 \neq 0$.

but $x^2+1 > 0$ for all values of x

because $x^2 \geq 0$ and $1 > 0$

Therefore $x^2+1 \neq 0$ for all values of x

Then $D_g =]-\infty; +\infty[$

III. Derivative

- If f is the function defined by $f(x) = \frac{P(x)}{q(x)}$ then its derivative function $f'(x)$ is defined by

$$f'(x) = \frac{P'(x) \times q(x) - q'(x) \times P(x)}{(q(x))^2}$$

Example

- Let f be the function defined by $f(x) = \frac{x^2+2x-3}{2x-1}$
- $$\begin{aligned} f'(x) &= \frac{(x^2+2x-3)' \times (2x-1) - (2x-1)' \times (x^2+2x-3)}{(2x-1)^2} \\ &= \frac{(2x+2)(2x-1) - 2(x^2+2x-3)}{(2x-1)^2} \\ &= \frac{4x^2 - 2x + 4x - 2 - 2x^2 - 4x + 6}{(2x-1)^2} \\ &= \frac{2x^2 - 2x + 4}{(2x-1)^2} \end{aligned}$$

IV. Center of symmetry

- Let f be a function defined over D_f and designate by (C) its representative curve in an orthonormal system $(O; \vec{i}; \vec{j})$
- A point $I(a; b)$ is said to be center of symmetry of (C) if :
 - 1) D_f is centered at a which means that
for all $x \in D_f$ we have $a - x \in D_f$ and $a + x \in D_f$
 - 2) $f(2a - x) + f(x) = 2b$

Example

- Let f be the function defined over $] -\infty; 1[\cup] 1; +\infty[$ by $f(x) = \frac{x^2}{x-1}$

Designate by (C_f) its representative curve in an orthonormal system $(o, \vec{i}, \vec{j})$

The point $I(1;2)$ is a center of symmetry of (C_f) because

- $] -\infty; 1[\cup] 1; +\infty[$ is centered at $x_I = 1$.
- For all $x \in] -\infty; 1[\cup] 1; +\infty[$ we have

$$f(2a - x) + f(x) = f(2 \times 1 - x) + f(x) = f(2 - x) + f(x) = \frac{(2 - x)^2}{2 - x - 1} + \frac{x^2}{x - 1}$$

$$\begin{aligned} &= \frac{4 - 4x + x^2}{1 - x} + \frac{x^2}{x - 1} \\ &= \frac{4 - 4x + x^2}{-(x - 1)} + \frac{x^2}{x - 1} \\ &= \frac{-4 + 4x - x^2}{x - 1} + \frac{x^2}{x - 1} \\ &= \frac{4(x - 1)}{(x - 1)} = 4 = 2b \end{aligned}$$

Oblique asymptote

- Let f be a rational function defined over D_f and designate by (C) its representative curve in an orthonormal system $(O; \vec{i}; \vec{j})$

A straight line (d) of equation $y = ax + b$ is said to be an oblique asymptote to (C) at $-\infty$

$$\text{If } \lim_{x \rightarrow -\infty} f(x) - (ax + b) = 0$$

(d) is said to be an oblique asymptote to (C) at $+\infty$ if

$$\lim_{x \rightarrow +\infty} f(x) - (ax + b) = 0$$

Example

- Let f be the function defined over $] -\infty; 1[\cup] 1; +\infty[$ by $f(x) = \frac{x^2}{x-1}$
- Designate by (C_f) its representative curve in an orthonormal system $(O, \vec{i}, \vec{j})$
- The straight line (d): $y = x + 1$ is an oblique asymptote to (C_f) at $-\infty$ and at $+\infty$ because

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) - (x + 1) &= \lim_{x \rightarrow \pm\infty} \frac{x^2}{x-1} - (x + 1) \\ &= \lim_{x \rightarrow \pm\infty} \frac{x^2 - (x + 1)(x - 1)}{x - 1}\end{aligned}$$

$$\begin{aligned}&= \lim_{x \rightarrow \pm\infty} \frac{x^2 - (x^2 - 1)}{x - 1} \\ &= \lim_{x \rightarrow \pm\infty} \frac{x^2 - x^2 + 1}{x - 1} \\ &= \lim_{x \rightarrow \pm\infty} \frac{1}{x - 1} = \frac{1}{\pm\infty} = 0\end{aligned}$$

Relative position of a curve of a function f and a straight line .

- Let f be a function defined over D_f and designate by (C) its representative curve in an orthonormal system $(O; \vec{i}; \vec{j})$
- Let $(d) : y = ax + b$ be a straight line.
- The relative position of (C) and (d) depends on the sign of $f(x) - y$
 - 1) For the values of x such that $f(x) - y = 0$ we say that (C) intersect (d) .
 - 2) For the values of x such that $f(x) - y < 0$ we say that (C) is below (d) .
 - 3) For the values of x such that $f(x) - y > 0$ we say that (C) is above (d) .

Example

- Let f be the function defined over $] -\infty; 1[\cup] 1; +\infty[$ by
$$f(x) = \frac{x^2}{x-1}$$
- Designate by (C_f) its representative curve in an orthonormal system $(o, \vec{i}, \vec{j})$
- The straight line $(d): y = x + 1$ is an oblique asymptote to (C_f) at $-\infty$ and at $+\infty$ proved
- To study the relative position of (C) and (d) we start by calculating $f(x) - y$
- $$f(x) - y = f(x) - (x + 1) = \frac{x^2}{x-1} - (x + 1) = \frac{x^2 - (x+1)(x-1)}{x-1} = \frac{x^2 - (x^2 - 1)}{x-1} = \frac{1}{x-1}$$
- Then we construct a table of sign of $f(x) - y$ as follow

- $f(x) - y = \frac{1}{x-1}$ is defined when $x - 1 \neq 0 \rightarrow x \neq 1$

x	$-\infty$	1	$+\infty$
$f(x) - y$	-		+
Relative position of (C) and (d)	(C) is below (d)		(C) is above (d)